

Milestone 2 Report: Object Interaction and Control Logic

Data Cleaning and Transformation Pipeline

1. Introduction

Following the foundation established in Milestone 1, this stage enhances the Data Cleaning and Transformation Pipeline by introducing object interaction and control logic.

While the previous milestone focused on building independent components in a linear pipeline, this milestone transforms the system into a more intelligent structure where components communicate and the system can make decisions during execution.

2. System Enhancement Overview

The system remains a Data Cleaning and Transformation Pipeline, but now includes interaction between objects, decision-making capabilities, and centralized coordination of system flow.

These enhancements allow the system to behave more intelligently rather than following a fixed sequence of operations.

3. Object Interaction

In this milestone, objects no longer operate independently. Instead, they exchange information to influence system behavior.

The Cleaner processes raw data and generates cleaned data and a report containing information such as the number of missing values.

The PipelineController receives this report and uses it to decide the next step.

The Transformer is executed conditionally based on the decision made by the controller.

This establishes a communication flow between objects, enabling coordinated processing.

4. Control Logic

Control logic is introduced to allow the system to make decisions based on data conditions.

If the number of missing values is greater than a defined threshold, transformation is skipped. Otherwise, transformation is applied.

This ensures that the system adapts its behavior depending on the quality of the input data.

5. Updated System Design

A new object, PipelineController, is introduced. It acts as the central coordinator of the system, manages execution flow, applies decision-making rules, and facilitates communication between objects.

The updated object structure includes DataSource, Cleaner, Transformer, OutputHandler, and PipelineController.

6. System Flow

DataSource → Cleaner → Controller → (Transform or Skip) → OutputHandler

Data is obtained from the DataSource, processed by the Cleaner, evaluated by the Controller, and then either transformed or skipped before output.

7. Prototype Implementation

```
class DataSource:
    def get_data(self):
        return [10, None, 20, None, None, 40]

class Cleaner:
    def clean(self, data):
        cleaned = [x for x in data if x is not None]
        missing_count = data.count(None)
        report = {"missing_count": missing_count}
        return cleaned, report

class Transformer:
    def transform(self, data):
        return [x * 2 for x in data]

class OutputHandler:
    def output(self, data):
        print("Final Output:", data)

class PipelineController:
    def __init__(self):
        self.data_source = DataSource()
        self.cleaner = Cleaner()
        self.transformer = Transformer()
        self.output_handler = OutputHandler()

    def run(self):
        data = self.data_source.get_data()
        cleaned_data, report = self.cleaner.clean(data)

        if report["missing_count"] > 2:
            print("Too many missing values. Skipping transformation.")
            final_data = cleaned_data
```

```
        else:
            final_data = self.transformer.transform(cleaned_data)

        self.output_handler.output(final_data)

    controller = PipelineController()
    controller.run()
```

8. Explanation of the Prototype

The DataSource provides raw data containing missing values. The Cleaner removes missing values and generates a report.

The PipelineController evaluates the report and applies control logic.

If missing values exceed the threshold, transformation is skipped; otherwise, the Transformer processes the data.

Finally, the OutputHandler displays the result.

This demonstrates interaction between objects, centralized decision-making, and conditional execution.

9. Conclusion

Milestone 2 successfully extends the system from a static pipeline to an interactive and controlled system.

The system now supports communication between components, applies logical rules during execution, and uses a controller to manage workflow.

These improvements prepare the system for the next milestone involving multi-object interaction and dynamic behavior.